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Introduction to Multi-Cloud Environments

This section explains why companies need a proper **plan (scaffold)** before starting to use cloud services such as Azure, AWS, or GCP. Many organizations begin deploying resources directly in the cloud without planning, which later leads to problems such as high costs, poor security, and lack of control.

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Think of it like building a data center. You would never start building a physical data center by randomly buying servers and cables. Similarly, cloud infrastructure also requires planning and management.

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1. Cloud Adoption Frameworks

Major cloud providers have created **Cloud Adoption Frameworks** to help organizations adopt cloud services in a structured way. These frameworks provide:

- Best practices
- Security guidelines
- Cost management strategies
- Governance policies
- Operational procedures

The frameworks from Azure, AWS, and GCP may use different terminology, but they all aim to help businesses stay in control of their cloud environments.

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What is a Scaffold?

A **scaffold** is a support structure used in construction. Microsoft adopted this term to describe the foundation used to build and manage cloud environments. Unlike a temporary construction scaffold, a cloud scaffold remains permanently and serves as the base for cloud operations.

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Key Idea

Even though you are using public cloud services, you are still essentially building a data center and should manage it with the same discipline.

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2. Hyperautomation in Multi-Cloud

Organizations often automate individual components separately. For example:

- Azure scripts
- AWS scripts
- GCP workflows

This creates multiple tools, workflows, and maintenance efforts. Instead, **Hyperautomation** aims to automate all business processes using a single integrated automation platform.

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Benefits of Hyperautomation

- Reduces repetitive manual work
- Reduces human errors
- Centralizes automation
- Improves efficiency
- Simplifies management across clouds

The future of hyperautomation is closely linked with **AIOps (Artificial Intelligence for IT Operations)**, where AI helps manage and optimize cloud operations automatically.

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3. Identities and Roles in the Cloud

Every user, application, virtual machine, or service in the cloud has an **identity**. Managing identities involves two major tasks:

A. Authentication

Authentication answers:

"Who are you?"

It verifies whether a user or resource is genuine.

Most organizations use **Active Directory (AD)** as the central identity store. AD stores information about:

- Employees
- Computers
- Servers
- Applications

Cloud resources must often be connected to the organization's domain to function properly. This is called **domain joining**.

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B. Authorization

Authorization answers:

"What are you allowed to do?"

Even after authentication, users should only access resources they are permitted to use. Authorization controls permissions and access rights.

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4. Active Directory Federation

Organizations often want their cloud resources to use the same identities as their on-premises systems.

Example:

Suppose a company owns the domain:

myfavdogbiscuit.com

The company can connect this domain to Azure, AWS, or GCP and allow cloud resources to become part of the organization's identity system. This connection is established using **Domain Controllers**.

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This process is known as **AD Federation**.

Benefits

- Single identity system
- Easier management
- Centralized authentication
- Consistent access control across multiple clouds

5. Authentication Technologies

Kerberos

Active Directory uses **Kerberos** to verify identities securely.

All major cloud providers support:

- Active Directory
- LDAP (Lightweight Directory Access Protocol)
- Kerberos

This allows integration between enterprise directories and cloud platforms.

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6. External User Access

Sometimes external users need access to company resources.

Cloud providers offer identity services for this:

Cloud Provider	External Identity Service
Azure	Azure AD B2B
AWS	Cognito
GCP	Cloud Identity

These services allow secure access for users who are not part of the organization's primary Active Directory.

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7. Multi-Factor Authentication (MFA)

For additional security, users should use **Multi-Factor Authentication (MFA)**.

MFA requires more than one verification method:

1. Password
2. OTP on mobile
3. Biometric verification

This significantly improves security.

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8. Role-Based Access Control (RBAC)

Once identities are verified, permissions must be assigned.

This is done through:

Cloud Provider	Access Control System
Azure	RBAC
AWS	IAM
GCP	Cloud Identity & IAM

RBAC allows administrators to:

- Create roles
- Assign permissions
- Group users
- Control resource access

For example:

- Developer → Deploy applications
- Database Admin → Manage databases
- Security Admin → Manage security policies

9. AD vs Azure AD

Many people think AD and Azure AD are the same, but they are different.

Active Directory (AD)

- Traditional directory service
- Uses Kerberos
- Stores enterprise identities
- Usually runs on company infrastructure

Azure AD

- Cloud-native identity provider
- Used within Azure
- Does not directly provide Kerberos authentication
- Relies on AD for authentication

AD remains the **single source of truth** for identity management across multiple clouds.

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10. Service Design and Governance Model

A multi-cloud service design should include:

1. Requirements

Define:

- Business objectives
- Service targets
- Cloud services required

2. Identity and Access Management

Define:

- Authentication methods
- Authorization policies
- RBAC roles
- MFA requirements

3. Governance

Define:

- Standards
- Policies
- Compliance requirements

4. Cost Management

Monitor:

- Cloud spending
- Budgets
- Resource utilization

5. Security

Protect:

- Data
- Applications
- Cloud resources
- User identities

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11. Service Level Agreements (SLAs)

Cloud providers publish SLAs that define:

- Availability guarantees
- Performance commitments
- Service reliability

Examples:

- Azure SLA
- AWS SLA
- GCP SLA

These SLAs only cover services provided by the cloud platform itself. They do **not** cover applications that a company builds on top of those services.

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10-Mark Exam Summary

Multi-cloud environments require proper planning through cloud adoption frameworks and scaffolding. These frameworks help organizations manage costs, governance, security, and operations. Identity management is achieved using Active Directory, authentication, authorization, MFA, and RBAC. Cloud providers support federation with enterprise directories, enabling centralized identity management. Hyperautomation integrates automation across multiple cloud platforms, reducing manual effort and human error. Finally, a service design and governance model should include requirements, security, identity management, governance policies, cost management, and SLA considerations to ensure efficient and secure multi-cloud operations.

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Managing the Enterprise Cloud Architecture (Detailed Explanation for 10 Marks)

This chapter explains how organizations should manage and maintain their cloud environments after adopting a multi-cloud strategy. Once a company starts using multiple cloud providers such as AWS, Azure, and GCP, it needs a structured architecture to ensure security, scalability, reliability, and governance.

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1. Enterprise Architecture and TOGAF

To manage a multi-cloud environment effectively, organizations use **Enterprise Architecture (EA)** frameworks.

One of the most popular frameworks is **TOGAF (The Open Group Architecture Framework)**.

TOGAF uses the **Architecture Development Method (ADM)**, which helps organizations design and manage their architecture systematically. Before creating the actual architecture, two important steps are performed:

1. Preliminary Phase
2. Architecture Vision

These phases establish the architecture principles that guide all future architectural decisions.

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2. Architecture Principles

Architecture principles are guidelines that help organizations achieve their business goals.

A principle should:

- Support business objectives
- Be clear and unambiguous
- Lead toward practical solutions
- Remain consistent over time

For example:

✗ "Cloud-first" is not a business goal.

✓ "Improve flexibility and reduce deployment time" is a business goal.

The architecture principles should always help the business achieve its objectives.

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Common Principle Categories

Organizations generally define principles in the following areas:

1. Business
2. Security and Compliance
3. Data
4. Applications
5. Infrastructure and Technology
6. Usability
7. Processes

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3. Quality Attributes in Architecture

Modern cloud environments change continuously because of:

- DevOps
- Continuous deployment
- Customer demands
- Rapid innovation

To manage these changes, Continuous Architecture defines several **Quality Attributes**.

These attributes determine whether the architecture is effective and sustainable.

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4. Operability

Operability means keeping systems running efficiently and securely.

It includes:

Monitoring

Monitoring checks:

- Whether systems are healthy
- Whether services are functioning correctly
- Whether errors occur

Logging

Logs record:

- User activities
- Errors
- Security events

Logs are important for:

- Troubleshooting
- Audits
- Compliance requirements

End-to-End Monitoring

Instead of monitoring individual components separately, organizations monitor the complete service chain from the user's perspective.

Benefits:

- Faster issue detection
- Better performance analysis
- Improved user experience

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5. Performance

Performance measures how quickly systems respond.

A system may technically work, but if users experience delays, it is considered poor performance.

Performance monitoring helps identify:

- Slow servers
- Network bottlenecks
- Database delays
- Application inefficiencies

Many organizations use monitoring agents that continuously measure response times and system health.

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6. Configurability

Cloud systems should be easy to configure and modify.

Configuration determines how a resource behaves.

Examples:

- CPU allocation
- Memory settings
- Disk sizes
- Security settings
- User permissions
- Operating system parameters

Instead of manually configuring every new resource, organizations use automated configuration management tools.

Examples:

- Desired State Configuration (DSC) in Microsoft environments
- AWS Systems Manager
- Built-in GCP configuration services

This ensures that every deployed resource follows company standards automatically.

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7. Discoverability

Discoverability means that services and resources can locate and communicate with each other automatically.

This is especially important in **Microservices Architecture**.

For example:

An e-commerce application may have:

- Payment Service
- Inventory Service
- Order Service

These services must automatically find each other even if server IP addresses change.

This is achieved through:

- Service Discovery

- Service Registries

Benefits:

- Dynamic communication
- Easier scaling
- Better reliability

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8. Security

Security focuses on protecting the most valuable asset:

Data

Architecture should protect:

- Data confidentiality
- Data integrity
- Data availability

Security architecture includes:

- Threat prevention
- Access controls
- Monitoring
- Security policies
- Alert management

The objective is to prevent unauthorized access and protect applications and data from attacks.

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9. Scalability

One of the biggest advantages of cloud computing is scalability.

Scalability allows resources to increase or decrease based on demand.

Horizontal Scaling (Scale Out)

Add more servers.

Example:

1 Web Server → 5 Web Servers

Vertical Scaling (Scale Up)

Increase resources of an existing server.

Example:

- 4 CPU → 8 CPU
- 8 GB RAM → 32 GB RAM

Scale Down

Remove unnecessary resources to reduce costs.

Benefits:

- Flexibility
- Cost savings
- Improved performance

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Autoscaling

Cloud platforms can automatically adjust resources.

Autoscaling requires:

- Defined thresholds
- Scaling rules
- Cost awareness

For example:

If CPU usage exceeds 80%, automatically add another server.

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Partitioning

Large systems are divided into smaller sections called partitions.

Benefits:

- Easier management
- Better performance
- Reduced contention
- Improved scalability

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10. Robustness and Availability

Cloud providers offer highly available infrastructure, but outages can still occur.

Therefore, systems must be designed for resilience.

Robustness includes:

- Availability
- Recovery
- Accessibility
- Data retention

Architects must ensure that:

- Compute systems remain available
- Applications remain available
- Data remains available

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11. RPO and RTO

Recovery Point Objective (RPO)

Maximum acceptable amount of data loss.

Example:

RPO = 1 hour

The business can tolerate losing at most 1 hour of data.

Recovery Time Objective (RTO)

Maximum acceptable downtime.

Example:

RTO = 30 minutes

The system must be restored within 30 minutes.

These values determine backup and disaster recovery strategies.

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12. Backup and Recovery

Organizations should:

- Take regular backups
- Use snapshots
- Use incremental backups
- Test restoration procedures

A backup is only useful if it can be restored successfully.

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13. Portability

Portability means applications can move between cloud providers without major changes.

Example:

An application running on AWS should also be able to run on Azure or GCP.

Benefits:

- Avoid vendor lock-in
- Greater flexibility
- Easier migration

Container technologies like:

- Kubernetes
- Docker

help improve portability.

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14. Usability

Usability focuses on the user experience.

Applications should be:

- Easy to access
- Easy to navigate
- Consistent across platforms

Architecture decisions affecting usability include:

- Network design
- Internet access
- DMZ configuration
- Routing

- Application accessibility

Users should not notice whether application components run on Azure, AWS, or GCP.

This concept is called **Technology Transparency**.

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15. Open Agile Architecture (O-AA)

TOGAF is powerful but relatively static.

To support modern agile businesses, The Open Group introduced:

Open Agile Architecture (O-AA)

Key features:

- Focuses on business agility
- Supports digital transformation
- Places customers at the center
- Adapts continuously to changing customer needs

O-AA considers customer interactions through:

- Websites
- Mobile apps
- Social media
- Contact centers
- IoT devices

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10-Mark Exam Conclusion

Enterprise cloud architecture provides a structured approach for managing multi-cloud environments. Using frameworks such as TOGAF and Open Agile Architecture, organizations define architecture principles and quality attributes that guide architectural decisions. Important quality attributes include operability, performance, configurability, discoverability, security, scalability, robustness, portability, and usability. Features such as autoscaling, monitoring, backup strategies, RPO, RTO, and service discovery help ensure reliability and business continuity. By following these principles, organizations can build secure, scalable, resilient, and agile multi-cloud environments that align with business objectives.

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Defining Principles from Use Cases (Detailed Explanation for 10 Marks)

Before designing any cloud architecture, architects must first understand the **business use case**. A business use case describes the sequence of activities a business performs to deliver a product or service to customers while generating profit. It explains how value is created for customers and how the organization achieves its business goals. From these use cases, architects derive architectural principles that guide the design of the multi-cloud environment.

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1. Business Use Cases

A **business use case** defines:

- What product or service is delivered
- How it is delivered
- Who receives it
- How value is created
- How profit is generated

For example:

An online shopping company may have the use case:

1. Customer visits website.
2. Customer places an order.
3. Payment is processed.
4. Product is shipped.
5. Customer receives the product.

This workflow becomes the basis for designing the supporting IT architecture.

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2. Business Principles

Business principles are derived from business goals and strategies.

These principles guide all future architectural decisions and ensure that technology supports business objectives.

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Common Business Goals

Organizations may aim to:

- Respond faster to customers
- Launch products faster (Time to Market)

- Improve product quality
- Improve employee engagement
- Achieve digital transformation
- Increase revenue and profit
- Control operational costs

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SMART Goals

Business goals should follow the **SMART** principle:

Letter	Meaning
S	Specific
M	Measurable
A	Attainable
R	Relevant
T	Timely

Example:

"Launch the web shop for Product X in North America on June 1."

This goal is:

- Specific
- Measurable
- Achievable
- Relevant
- Time-bound

Therefore, it is a SMART goal.

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Importance of Business Principles

Business principles:

- Support strategic decisions
- Align IT with business objectives
- Serve as prerequisites for architecture design
- Guide future cloud adoption decisions

In TOGAF, defining business principles is part of the **Business Architecture Phase (Phase B)** of the ADM cycle.

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3. Security and Compliance Principles

Security is one of the most important architectural domains.

Two commonly used principles are:

A. Zero Trust

Zero Trust means:

Never trust anything automatically.

Every:

- User
- Device
- Application
- Service

must continuously prove its identity and authorization.

Key features:

- Continuous monitoring
- Micro-segmentation
- Strict access control
- Assume breach mentality

Instead of asking:

"Can someone attack us?"

Zero Trust assumes:

"An attack has already happened; how do we limit the damage?"

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B. Security by Design

Security should be built into systems from the beginning rather than added later.

Examples:

- Secure coding practices
- Network segmentation
- System hardening
- Encryption
- Authentication mechanisms

This ensures every component is secure by default.

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CIS Framework

A commonly used security framework is the **Center for Internet Security (CIS)** framework.

It provides:

- 20 Critical Security Controls
- Best practices for protecting systems
- Security recommendations across multiple IT layers

Example:

Data Protection (Control 13)

CIS recommends:

- Encrypting data at rest
- Encrypting data in transit
- Securing encryption keys

The architect decides which encryption methods and security technologies should be used.

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4. Compliance Principles

Organizations must comply with laws and regulations.

Examples include:

- Privacy laws
- Industry regulations
- Government regulations

One important example is:

GDPR (General Data Protection Regulation)

A principle may state:

"The architecture must comply with GDPR."

To achieve this, architects may implement:

- Database security
- Data encryption
- Authentication mechanisms
- Authorization controls
- Privacy protection measures

Compliance becomes more complex in multi-cloud environments because data may be stored across multiple cloud providers and geographical regions.

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5. Data Principles

Data is one of the most valuable assets in a business.

Five important data principles are:

1. Accuracy

Data should be:

- Correct
- Complete
- Reliable

Inaccurate data may lead to poor business decisions.

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2. Relevance

Data should directly support the business problem being solved.

Unnecessary data:

- Increases complexity
- Creates noise
- Reduces analysis quality

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3. Timeliness

Data must be current and up-to-date.

Outdated information may result in:

- Incorrect reports
- Poor forecasting
- Wrong decisions

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4. Consistency

Data should have:

- Consistent formats
- Standard definitions
- Uniform measurement units

Consistency allows easy comparison and integration of information from different sources.

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5. Privacy and Security

Sensitive information must be protected from:

- Unauthorized access
- Data breaches
- Data leaks

Examples:

- Personal information
- Financial records
- Healthcare data

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6. Systems of Record

A **System of Record** stores business data.

Examples:

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- Databases
- ERP systems
- CRM systems
- Data Lakes

Characteristics:

- Store authoritative data
- Act as the source of truth
- Maintain historical records

Cloud platforms are ideal for large-scale data storage because they can host huge data repositories and data lakes.

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7. Systems of Engagement

A **System of Engagement** interacts with users and accesses data.

Examples:

- Email systems
- Collaboration platforms
- Mobile applications
- Content Management Systems
- IoT devices

Functions:

- Collect data
- Display data
- Exchange data with Systems of Record

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8. Systems of Intelligence

As organizations become data-driven, a new layer is emerging:

Systems of Intelligence

These systems use:

- Artificial Intelligence (AI)
- Machine Learning (ML)

- Advanced Analytics

Purpose:

- Analyze large datasets
- Generate predictions
- Support business decisions
- Automate actions

Examples:

Cloud Provider	AI Service
Azure	Azure ML
AWS	SageMaker
GCP	Cloud AI

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Architecture Layers

Systems of Engagement
↓
Systems of Record
↓
Systems of Intelligence

Example

Online Shopping Platform:

Systems of Engagement

- Website
- Mobile App

Systems of Record

- Customer Database
- Product Database

Systems of Intelligence

- Product Recommendation Engine
- Fraud Detection System
- Sales Forecasting

10-Mark Exam Conclusion

Architecture principles are derived from business use cases and guide the design of multi-cloud environments. Business principles support organizational goals through SMART objectives, while security principles such as Zero Trust and Security by Design ensure strong protection. Compliance principles help organizations meet regulations such as GDPR. Data principles focus on accuracy, relevance, timeliness, consistency, and security. Modern architectures organize data using Systems of Record and Systems of Engagement, while emerging Systems of Intelligence use AI and machine learning to generate business insights. Together, these principles create secure, scalable, compliant, and business-aligned multi-cloud architectures.

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Application principles

Data doesn't stand on its own. If we look at TOGAF once more, we'll see that data and applications are grouped into one architectural phase, known as phase C, which is the information systems architecture. In modern applications, one of the main principles of applications is that they have a data-driven approach, following the recommendation of Steven Spewak's enterprise architecture planning. Spewak published his book Enterprise Architecture Planning in 1992, but his approach is still very relevant, even—and perhaps even more—in multi-cloud environments.

Also mentioned in Spewak's work: the business mission is the most important driver in any architecture. That mission is data-driven; enterprises make decisions based on data, and for that reason, data needs to be relevant, but also accessed and usable. These latter principles are related to the applications disclosing the data to the business. In other words, applications need to safeguard the quality of the data, make data accessible, and ensure that data can be used. Of course, there can be—and there is—a lot of debate regarding, for instance, the accessibility of data. The sole relevant principle for an application architecture is that it makes data accessible.

To whom and on what conditions are both security principles.

In multi-cloud, the storage data changes, but also the format of applications. Modern applications are usually not monolithic or client-server-based these days, although enterprises can still have a large base of applications with legacy architectures. Cloud-native apps are defined with roles and functions and build on the principles of code-based modularity and the use of microservices. These apps communicate with other apps using APIs or even triggers that call specific functions in other apps. These apps don't even have to run on the same platform; they can be hosted anywhere. Some architects tend to think that monolithic applications on mainframes are complex, so use that as a guideline to figure out how complex apps in multi-cloud can get.

However, a lot of architectural principles for applications are as valid as ever. The technology might change, but the functionality of an application is still to support businesses when it comes to rendering data, making it accessible, and ensuring that the data is usable.

Today, popular principles for applications are taking the specific characteristics of cloud-native technology into consideration. Modern apps should be enabled for mobility, be platform-independent using open standards, support interoperability, and be scalable. Apps should enable users to work with them at any time, anywhere.

Chapter 5 121

One crucial topic is the fact that the requirements for applications change at almost the speed of light: users demand more and more from apps, so they have to be designed in an extremely agile way so that they can adopt changes incredibly fast in development pipelines. Cloud technology does support this: code can easily be adapted. But this does require that the applications are well-designed and documented, including in runbooks.

Infrastructure and technology principles

Finally, we get to the real technology: machines, wires, nuts, and bolts. Here, we're talking about virtual nuts and bolts. Since data is stored in many places in our multi-cloud environment and applications are built to be cloud-native, the underlying infrastructure needs to support this. This is phase D in TOGAF, the phase in architecture development where we create the target technology

architecture, which comprises the platform's location, the network topology, the infrastructure components that we will be using for specific applications and data stores, and the system interdependencies. In multi-cloud, this starts with drafting the landing zone: the platform where our applications and data will land.

One of the pitfalls of this is that architects create long, extensive lists with principles that infrastructure and technology should adhere to, all the way up to defining the products that will be used as a technology standard. However, a catalog with products is part of a portfolio. Principles should be generic and guiding, not constraining. In other words, a list of technology standards and products is not a principle. A general principle could be about bleeding edge technology: a new, non-proven, experimental technology that imposes a risk when deployed in an environment because it's still unstable and unreliable.

Other important principles for infrastructure can be that it should be scalable (scale out, up, and down) and that it must allow micro-segmentation. We've already talked about the Twelve-Factor App, which sets out specific requirements for the infrastructure. These can be used as principles. The principles for the Twelve-Factor App were set out in 2005, but as we already concluded in Chapter 2, Collecting Business Requirements, they are still very accurate and relevant.

The Twelve-Factor App sets the following three major requirements for infrastructure:

- The app is portable between different platforms, meaning that the app is platform-agnostic and does not rely on a specific server or system's settings.
- There's little to no difference between the development stage and the production stage of the app so continuous development and deployment are enabled. The platform that the app is deployed on should support this (meaning that everything is basically code-based).

122 Managing the Enterprise Cloud Architecture

- The app supports scaling up without significant changes needing to be made to the architecture of the app.

In the next section, we will discuss the principles for usability and processes. We will also touch upon the transition and transformation to cloud environments.

Principles for processes

The last group of principles is concerned with processes. This is not about the IT System Management (ITSM) processes, but about the processes of deployment and automation in multi-cloud.

One of the principles in multi-cloud is that we will automate as much as we can. This means that we will have to define all the tasks that we would typically do manually in an automated workflow. If we have a code-only principle defined, then we can subsequently set a principle that states that we must work from the code base or master branch. If we fork the code and we do have to make changes to it, then a principle is that altered code can only be committed back to the master code if it's tested in an environment that is completely separated from acceptance and production. This is related to the life cycle process of our environment.

So, processes here focus more on our way of working. Today, a lot of companies are devoted to agile and DevOps. If that's the defined way of working, then it should be listed as a principle; for example, an organization embraces agility in its core processes. This can be done through the Scaled Agile Framework (SAFe) or the Spotify model. Following that principle, a company should also define the teams, their work packages, and how epics, features, product backlogs, and so on are planned. However, that's not part of the principle anymore. That's a consequence of the principle.

As with all principles, the biggest pitfall is making principles too complex. Especially with processes, it's important to really stick to describing the principle and not the actual process.

We have discussed the architecture principles and why we are doing architecture. The next step is to define the components of the architecture. This is the topic of the next section.

Creating the architecture artifacts

The hierarchy in documents that cover the architecture starts with the enterprise architecture. It's

the first so-called architecture artifact. The enterprise architecture is followed by the high-level design and the low-level design, which covers the various components in the IT landscape. We will explore this in more detail in the following sections. Keep in mind that these sections are merely an introduction to the creation of these artifacts. You will find samples of these artifacts at <https://publications.opengroup.org/i093>, where you can download a ZIP file containing relevant templates

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Application Principles and Infrastructure Principles (Detailed Explanation for 10 Marks)

1. Application Principles

Applications and data are closely connected. According to **TOGAF**, both belong to **Phase C – Information Systems Architecture**. Modern applications follow a **data-driven approach**, where business decisions are made using data. This concept was strongly emphasized by Steven Spewak in his book *Enterprise Architecture Planning* (1992), and it remains highly relevant in multi-cloud environments today.

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Role of Applications

Applications must:

- Protect data quality
- Make data accessible
- Ensure data is usable
- Support business operations

The primary purpose of an application is not merely processing information but enabling businesses to effectively use data for decision-making.

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Modern Cloud-Native Applications

Traditional applications were often:

- Monolithic
- Client-server based

Modern cloud-native applications are:

- Modular
- Based on microservices
- API-driven
- Event-driven

- Platform independent

Features include:

- Microservices architecture
- Communication through APIs
- Function triggers
- Distributed deployment across multiple clouds

Applications can run on different cloud platforms while still communicating with each other.

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Important Application Principles

Modern applications should:

1. Support mobility
2. Be platform-independent
3. Use open standards
4. Support interoperability
5. Be scalable
6. Be accessible anytime and anywhere

These principles ensure flexibility and compatibility across multiple cloud providers.

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Agility in Applications

Business requirements change rapidly.

Therefore applications must:

- Adapt quickly
- Support continuous development
- Support continuous deployment
- Be properly documented
- Include runbooks for operations

Cloud technologies enable rapid code changes, but applications must be well-designed to benefit from this agility.

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2. Infrastructure and Technology Principles

Infrastructure principles define how the underlying cloud environment should be designed and managed.

In TOGAF, this belongs to **Phase D – Technology Architecture**.

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Technology Architecture Covers

- Cloud platforms
- Landing zones
- Network topology
- Infrastructure components
- Data storage systems
- System dependencies

The landing zone serves as the foundation where applications and data are deployed in the cloud.

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Important Infrastructure Principles

Infrastructure should:

- Be scalable
- Support micro-segmentation
- Be secure
- Support cloud-native applications
- Allow automation

One common mistake is defining specific products as principles. Principles should remain generic and guide decisions rather than force particular technologies.

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Avoiding Bleeding-Edge Technology

A useful principle is avoiding unproven technologies.

Bleeding-edge technology refers to:

- Experimental solutions
- Unstable products
- Technologies lacking maturity

Using such technologies may introduce operational risks.

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3. The Twelve-Factor App Principles

The **Twelve-Factor App** provides important infrastructure guidelines for cloud-native applications.

Principle 1: Portability

Applications should run on different platforms without major modifications.

Benefits:

- Avoids vendor lock-in

- Simplifies migration
- Supports multi-cloud deployment

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Principle 2: Development and Production Consistency

There should be minimal differences between:

- Development environments
- Testing environments
- Production environments

This enables:

- Continuous Integration (CI)
- Continuous Deployment (CD)
- Faster releases

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Principle 3: Scalability

Applications should scale without requiring major architectural redesign.

Benefits:

- Better performance
- Increased flexibility
- Efficient resource utilization

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4. Process Principles

Process principles focus on how teams work and deploy solutions in multi-cloud environments.

These principles are concerned with:

- Automation
- Development workflows
- Deployment pipelines
- Agile practices

They are not traditional IT service management processes.

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Automation First

A major multi-cloud principle is:

Automate as much as possible.

Tasks that were previously performed manually should be converted into automated workflows.

Examples:

- Infrastructure deployment
- Application deployment
- Testing
- Monitoring
- Configuration management

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Code-Based Operations

Organizations often follow a **Code-Only Principle**:

- Infrastructure is managed as code
- Changes originate from the master branch
- Modifications require testing before deployment

This ensures consistency and reduces human errors.

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Agile and DevOps Principles

Many organizations adopt:

- Agile
- DevOps
- SAgile (Scaled Agile Framework)
- Spotify Model

A common principle is:

"The organization embraces agility in its core processes."

This influences team structures, planning, backlogs, epics, and feature development.

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Keep Principles Simple

One major recommendation is:

- Principles should be simple
- Principles should not describe detailed procedures
- Processes should be derived from principles

Overly complex principles create confusion and reduce effectiveness.

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5. Architecture Artifacts

After defining principles, architects create documentation called **Architecture Artifacts**.

The hierarchy is:

```
Enterprise Architecture
  ↓
High-Level Design (HLD)
  ↓
Low-Level Design (LLD)
```

Enterprise Architecture

Provides the overall organizational architecture and strategy.

High-Level Design (HLD)

Describes:

- Major components
- Interactions
- Architecture overview

Low-Level Design (LLD)

Provides detailed technical specifications for:

- Infrastructure
- Networks
- Applications
- Databases

These artifacts help ensure consistency and proper implementation across the multi-cloud environment.

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10-Mark Exam Conclusion

Application principles ensure that applications protect, expose, and enable the use of business data. Modern cloud-native applications use microservices, APIs, open standards, interoperability, scalability, and mobility. Infrastructure and technology principles define the cloud landing zone, network topology, scalability, security, and automation requirements. The Twelve-Factor App emphasizes portability, consistency across environments, and scalability. Process principles focus on automation, DevOps, Agile practices, and code-driven deployments. Together, these principles guide the creation of enterprise architecture, high-level designs, and low-level designs for successful multi-cloud implementations.

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